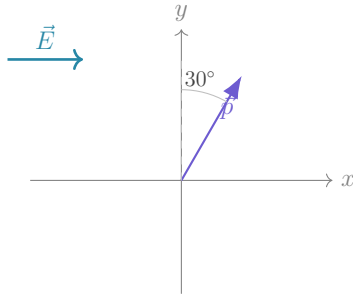


Electrostatics Review (completed) — Flex Day, Sep 30

Part A — Dipole: Torque, PE, and Motion

1. A uniform field $\vec{E} = 40 \text{ N/C}$ points in $+\hat{x}$. A dipole is made of two $\pm 3.0 \text{ nC}$ charges separated by $d = 2.0 \text{ mm}$, with its moment $\vec{p}$ tilted 30° to the right of the y -axis (i.e. 60° from $\vec{E}$).



(a) What is the torque on the dipole, as a vector (magnitude and direction)?
 $p = qd = (3.0 \times 10^{-9})(2.0 \times 10^{-3}) = 6.0 \times 10^{-12} \text{ C}\cdot\text{m}$.

$$\tau = pE \sin \theta = (6.0 \times 10^{-12})(40) \sin 60^\circ = 2.1 \times 10^{-10} \text{ N}\cdot\text{m}$$

Direction: [into the page](#) — $\vec{\tau} = \vec{p} \times \vec{E}$ points the way that *decreases* θ , rotating $\vec{p}$ toward alignment with $\vec{E}$.

(b) What is the dipole's potential energy in this orientation?

$$U = -pE \cos \theta = -(6.0 \times 10^{-12})(40) \cos 60^\circ = -1.2 \times 10^{-10} \text{ J}$$

(c) The dipole is released from rest and rotates until $\vec{p}$ aligns with $\vec{E}$ (i.e. $\theta \rightarrow 0$). Taking each charge's mass to be $m = 1.2 \times 10^{-14} \text{ kg}$, what speed does each charge have at the instant $\theta = 0$?
Hint: both charges move on a circle of the same radius about the center, so they share the same speed — the dipole's lost PE becomes the total KE of both charges together.

At $\theta = 0$: $U_0 = -pE = -2.4 \times 10^{-10} \text{ J}$. The PE lost is $\Delta U = U(60^\circ) - U(0^\circ) = (-1.2 \times 10^{-10}) - (-2.4 \times 10^{-10}) = 1.2 \times 10^{-10} \text{ J}$, which becomes the total KE of *both* charges together: $2 \times \frac{1}{2}mv^2 = mv^2 = \Delta U$.

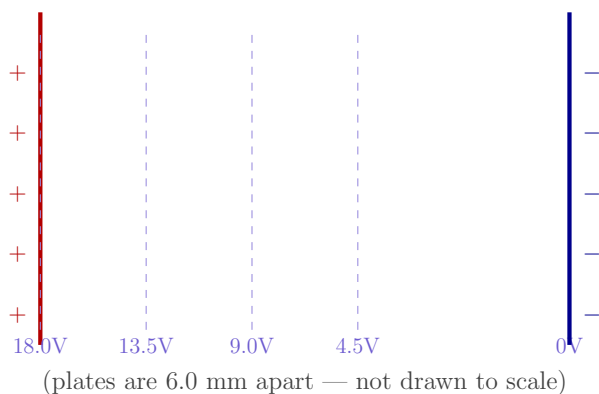
$$v = \sqrt{\frac{\Delta U}{m}} = \sqrt{\frac{1.2 \times 10^{-10}}{1.2 \times 10^{-14}}} = 100 \text{ m/s}$$

Part B — Even Steps in Potential, Uniform Field

2. A parallel-plate capacitor has plate separation 6.0 mm and a uniform field $E = 3000 \text{ N/C}$ between the plates, pointing from the + plate to the – plate.

(a) Draw the field lines between the plates (evenly spaced, pointing from + to –).

(b) Draw four equally-spaced equipotential lines between the plates, and label the potential of each one — take $V = 0$ at the negative plate.



$\Delta V_{\text{total}} = Ed = (3000)(0.0060) = 18.0 \text{ V}$, in four 4.5 V steps — the equipotentials are evenly spaced because E is constant, so equal physical steps give equal potential steps.

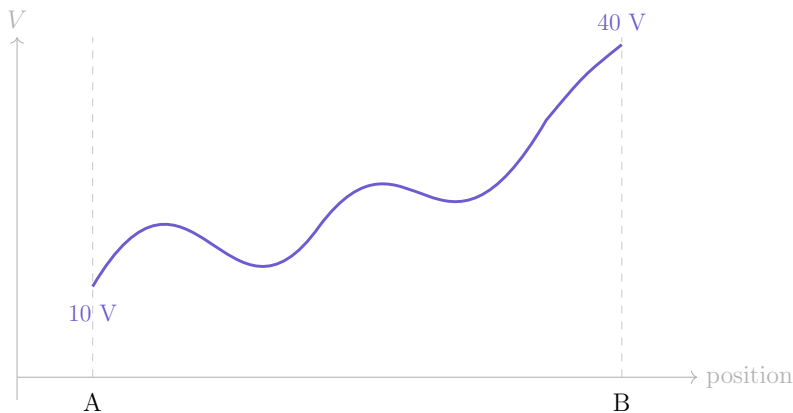
(c) A $+4.0 \text{ nC}$ charge moves from the positive plate to a point 1.5 mm away (one quarter of the way across). How much potential energy does it lose?

One quarter of the way corresponds to one 4.5 V step (V drops from 18.0 V to 13.5 V):

$$|\Delta U| = q|\Delta V| = (4.0 \times 10^{-9})(4.5) = 1.8 \times 10^{-8} \text{ J}$$

Part C — V Doesn't Have to Change Evenly to Know ΔU

3. A charge moves from point A, where $V_A = +10 \text{ V}$, to point B, where $V_B = +40 \text{ V}$ — but the potential along the way is *not* uniform: it rises and dips in between (like a bumpy hill, not a smooth ramp). Sketch a plausible bumpy $V(x)$ between A and B below, consistent with those two endpoint values.



Any bumpy curve is fine, as long as it starts at 10 V and ends at 40 V .

(a) A $+2.0 \text{ nC}$ charge moves from A to B along this bumpy path. What is ΔU ?

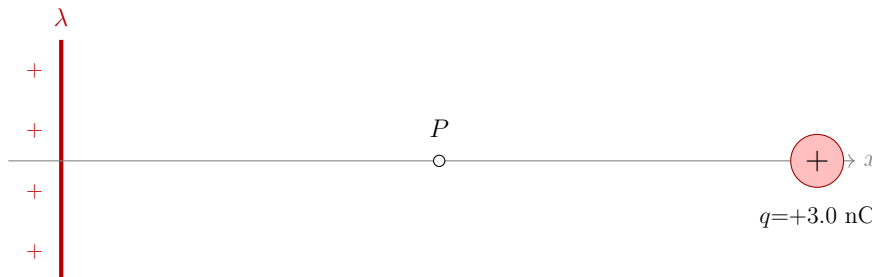
$$\Delta U = q\Delta V = (2.0 \times 10^{-9})(40 - 10) = 6.0 \times 10^{-8} \text{ J}$$

(b) Does the bumpiness of the path matter to your answer in (a)? Why or why not? (Think about the ball-on-a-hill analogy — does it matter if the hill between the start and end points is smooth or bumpy?)

No. $\Delta U = q\Delta V$ depends only on the potential at the two *endpoints*, never on the path between them — exactly like a ball rolling down a bumpy hill: only the starting and ending height set the change in gravitational PE, no matter how many ups and downs are in between.

Part D — Superposition with Continuous Distributions

4. An infinite line of charge with density $\lambda = +2.0 \times 10^{-8}$ C/m runs along the y -axis. A point charge $q = +3.0$ nC sits on the x -axis at $x = 10$ cm. Point P sits on the x -axis at $x = 5.0$ cm — between the line and the point charge.



(a) Find the field at P due to the line alone.

$$E_{\text{line}} = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2.0 \times 10^{-8}}{2\pi(8.85 \times 10^{-12})(0.050)} = 7.2 \times 10^3 \text{ N/C, in } +\hat{x}$$

(away from the line, since $\lambda > 0$).

(b) Find the field at P due to the point charge alone.

$$E_{\text{point}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = (8.99 \times 10^9) \frac{3.0 \times 10^{-9}}{(0.050)^2} = 1.08 \times 10^4 \text{ N/C, in } -\hat{x}$$

(toward the line, since P is to the left of the positive point charge, and the field points *away* from a positive charge — away from q at P means back toward the line).

(c) Add them as vectors (careful with directions!) to find the net field at P .

$$E_{\text{net}} = E_{\text{line}} - E_{\text{point}} = 7.2 \times 10^3 - 1.08 \times 10^4 = -3.6 \times 10^3 \text{ N/C (i.e. } 3.6 \times 10^3 \text{ N/C in } -\hat{x})$$

The point charge wins, even though it's the same distance away as the line — these two sources just don't produce the same field strength at that distance.

5. An infinite plane of charge with density $\eta = +1.0 \times 10^{-9}$ C/m² lies in the yz -plane. A point charge $q = +5.0$ nC sits on the x -axis at $x = 2.0$ cm, on the same side as point P at $x = 1.0$ cm.

(a) Find the field at P due to the plane alone, and due to the point charge alone.

$$E_{\text{plane}} = \frac{\eta}{2\epsilon_0} = \frac{1.0 \times 10^{-9}}{2(8.85 \times 10^{-12})} = 56 \text{ N/C, in } +\hat{x}$$

$$E_{\text{point}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = (8.99 \times 10^9) \frac{5.0 \times 10^{-9}}{(0.010)^2} = 4.5 \times 10^5 \text{ N/C, in } -\hat{x}$$

($r = 1.0$ cm from q at $x = 2.0$ cm to P at $x = 1.0$ cm; direction is $-\hat{x}$ since P is to the left of the positive point charge, so away-from- q at P points back toward the plane.)

(b) Add them as vectors to find the net field at P . Which source dominates, and does that surprise you given how differently each one's field falls off with distance?

$$E_{\text{net}} = E_{\text{plane}} - E_{\text{point}} = 56 - 4.5 \times 10^5 \approx -4.5 \times 10^5 \text{ N/C (i.e. } 4.5 \times 10^5 \text{ N/C in } -\hat{x})$$

The point charge dominates completely, even though an infinite plane sounds like it should be the bigger source. The plane's field never falls off with distance, but here P is extremely close to the point charge (1.0 cm) — close enough that $1/r^2$ wins easily over the plane's constant-but-modest field. Get far enough away and the ranking would flip: the plane's field stays exactly the same while the point charge's keeps shrinking.